

PACT Literature Review Protocol

Version 7 · 18 August 2026

Objective

Build a list of high-risk thinking tasks performed by clinicians in adult emergency medicine and primary care, drawn from the published literature. Expected output is roughly 20 to 30 tasks. That number is never given to any model or reviewer.

Approach

A bounded review of PubMed. A random sample of papers, screened and analysed with AI assistance, with people checking the work at four points. No existing PACT task list is used at any step.

Step 1. Search

Script: [01_search.py](#)

Papers must be about adult emergency medicine or primary care, and must involve at least one of:

- **How clinicians think.** Clinical reasoning, judgment, gestalt, uncertainty, cognitive load, situation awareness, named biases, missed and delayed diagnosis.
- **Where things go wrong.** Diagnostic and medical error, adverse events, malpractice claims, near misses, undertriage, failure to rescue.
- **Coordination and communication.** Handoff, care transition, referral, consultation, test result follow-up, shared decision making, goals of care, interruptions, communication failure, disposition and triage decisions, alert overrides.

Excluded: paediatrics, ICU, other specialties, nursing-only, education-only, case reports, editorials, and papers about patient thinking rather than clinician thinking. English, 2005 to 2026.

Before moving on: run the search, record how many papers each setting returns, and confirm the coordination terms return a usable number.

Step 2. Draw the sample

Script: [11_neutral_sample.py](#)

- 250 papers from emergency medicine, 250 from primary care
- Random draw; record the seed so it can be reproduced
- Number the papers in batches of 100 (needed in step 12)

Step 3. AI screening

Script: `03_screen.py`

Two models read all 500 titles and abstracts separately and decide whether each paper is in scope. A paper is in scope when it actually examines what a clinician notices, decides, interprets, or communicates. A paper that merely mentions a clinical decision in passing is not.

When a model is unsure, it includes. A wrongly kept paper costs almost nothing. A wrongly dropped one can remove a task from the final list.

Record which prompt file and which version of the code were used.

Step 4. Human check: did the AI keep the right papers?

300 papers, evenly split four ways:

	ED	Primary care
At least one model kept it	75	75
Both models dropped it	75	75

The dropped papers are the important half. They show what the AI was discarding.

Warm-up: all six reviewers read and discuss the same 10 abstracts first. These don't count toward anyone's total.

Who reads what: six blocks of 50, two reviewers per block, no two reviewers paired twice.

Block	Papers	Reviewers
1	1–50	R1, R2
2	51–100	R2, R3
3	101–150	R3, R4
4	151–200	R4, R5
5	201–250	R5, R6
6	251–300	R6, R1

Each reviewer reads 100. Every paper gets two independent reads. Rotating the pairs means a reviewer who is out of step with everyone else becomes visible.

Reviewer files show no task list.

Step 5. Sort out the disagreements

Script: `12_merge_human_reviews.py`

Where both reviewers agree, that stands. Where they disagree, a third person decides. Report agreement for each of the six pairs.

Step 6. Decide whether the AI screen can be trusted

Script: `07_metrics.py`

The number that matters: **of the papers a human would have kept, how many did the AI keep?**

Result	What we do with the other 200 papers
AI kept almost all of them	Use the AI screen as is
AI missed some	Keep anything either model flagged; a human checks the rest
AI missed a lot	The AI can't drop anything; it only sorts papers for human reading

The 300 were deliberately balanced rather than randomly drawn, so their numbers describe this check, not the literature. Don't quote them as rates.

Step 7. Pull the tasks out of the papers

Script: `06_extract.py`, run on every paper we keep.

For each paper, record:

- The name the paper gives a thinking task, kept only if the paper also defines it
- The paper's own definition
- An exact sentence from the paper as proof, checked automatically against the text
- Whether it's **a thinking process** (what the mind does) or **a clinical activity** (what the clinician does)
- Any named biases, recorded as ways a task fails rather than as tasks themselves
- The clinical situation, plus any error or frequency figure the paper gives

Labels too vague to test, like "clinical reasoning" on its own, are rejected.

Thinking processes go on to step 9. Clinical activities go into a separate list, reported alongside.

We pull the full PDF only when the abstract is too thin, when the paper looks like it holds something the abstract doesn't show, or when the supporting sentence can't be confirmed.

Step 8. Human check: are these tasks really in the papers?

Automatic, on everything: the script checks each quoted sentence word for word against the paper. Report how many failed.

By hand, 120 rows. Weighted toward tasks that appeared in only one or two papers, since that's where a made-up task can survive into the final list. Three questions each:

1. Is this the paper's wording, or did the AI supply the name?
2. Does the quote actually support it?
3. Is it a thinking process or a clinical activity?

One reviewer per row, with 30 rows read twice so agreement can be reported. Rows that fail question 1 or 2 are deleted before grouping.

This is the check that matters most. If bad rows reach the grouping step, the AI will build tidy, confident categories out of them, and step 11 won't catch it.

Step 9. Grouping, round one

Script: `08_cluster.py`, first pass.

Extraction leaves several hundred individual rows. Too many to hand over at once, so they go in batches of 50.

For each batch, the model proposes whatever tasks that batch supports, citing the papers. Batches can't see each other, so the same task will come back under different names from different batches. That's expected and not a problem. No target number is given.

Step 10. Grouping, round two

Script: `08_cluster.py`, second pass.

All the round-one proposals go to the model together. Now it can see everything side by side, so it combines the duplicates and keeps the genuinely different ones. It logs which proposals went into each surviving task.

This produces the candidate list.

Step 11. Human check: was the combining right?

Round two is where the list actually gets decided, and its mistakes leave no trace. Combining two different things deletes one of them permanently. Failing to combine two of the same thing inflates the count. Neither is visible in the output.

About 60 pairs, checked both ways:

- **Things it combined.** Two proposals with their quotes. Same task or different?
- **Things it kept apart.** State the boundary between them. If you can't, they should have been combined.

Two reviewers, with a named tiebreaker agreed in advance. **Unlike the other checks, this one changes the list.** What reviewers decide gets applied, and then the list is locked.

Step 12. Human check: do the groups hold together?

Done on the locked list.

Pull 10 items from each task, the same number regardless of how big the task is. Show each item next to three task names in random order: its own and two others. The reviewer picks the best fit.

If people pick the right one most of the time, the task is a real category. If they're picking at random, around a third of the time, the label isn't describing anything and that task is flagged as weak.

Separately: does the task name actually describe what's in it?

Agreed in advance: we report the results and change nothing. No task gets split, combined, or recalculated after this. Rewriting the groups based on this check would turn it into another round of group-making.

Step 13. Two summary checks

- **Where new tasks stopped appearing.** Plot new tasks found against batch number. This is the answer to why the sample is the size it is.
- **Stability.** Group each half of the papers separately and see whether you get roughly the same answer.

What comes out

1. The cognitive task list: each task with a definition, source papers, supporting quotes, and any error or frequency evidence
2. A separate clinical activity list
3. The saturation plot and the stability check

4. A validation report covering all four human checks

How to describe it

A bounded, targeted inductive review. Report the search terms and dates, how many papers were eligible, the sample size and seed, the 300-paper check and what it found, the six-reviewer setup, all four human checks, and how many full texts were pulled.

The task list was derived independently of any existing PACT task list. Don't call the review exhaustive, systematic, or scoping.

Limits

- PubMed only. It under-covers human factors and human-computer interaction work, most of which sits in psychology and computer science journals.
- A random sample of what the search returned, not all of it.
- The search terms decide what can be found. Anything with no vocabulary in our three term groups stays invisible.
- The 300-paper check was deliberately balanced, so its numbers aren't rates.
- Extraction and grouping are AI-generated. The four checks limit the error; they don't remove it.
- Abstracts leave out detail that appears in full text, and we pull full text selectively.
- The result is a candidate list drawn from the literature, not a systematic review.

Repository

github.com/Aschoeff613/PACT_Literature_Review

Record the prompt file and code version before running each step.